

Rounding Multi-Digit Numbers to Named Places

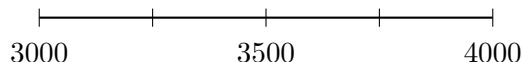
Rounding on a number line, rounding to a named place, estimating a sum or difference by rounding first, and rounding real counts and distances

Use the number line every time. Rounding to a named place means choosing between the two multiples of that place the number sits between.

- Find the two benchmarks — the multiple of the place just below the number and the one just above it.
- Mark the halfway point between them. It is always the lower benchmark plus half of the place.
- If the number lands past halfway, round up; before halfway, round down; exactly on halfway, round up.
- Every digit to the right of the named place becomes a zero. Nothing to the left of it changes, unless rounding up carries into it.

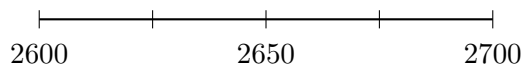
Numbers on this sheet are written without commas.

- 1.** Round 3847 to the nearest thousand. First write the two benchmarks and the halfway number, then mark 3847 on the line with an X.



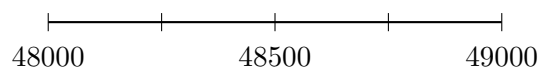
Answer: _____

- 2.** Round 2614 to the nearest hundred. Write the benchmark above the number and the halfway number, then mark 2614 on the line.



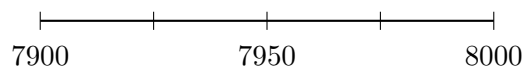
Answer: _____

- 3.** Round 48362 to the nearest thousand. The line already shows the lower benchmark; fill in the halfway number and the upper benchmark as you work.



Answer: _____

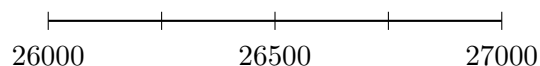
- 4.** Round 7952 to the nearest hundred. Watch what happens to the thousands digit when you round up.



Is 7952 before or after the halfway number? Say how you know.

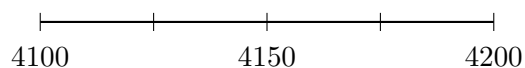
Answer: _____

- 5.** A stadium counted 26481 fans at a game. The newspaper prints the crowd size rounded to the nearest thousand. What number does the newspaper print?



Answer: _____

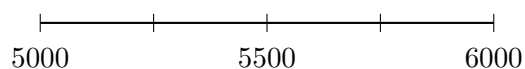
- 6.** Estimate $4182 + 2759$ by rounding each number to the nearest hundred first. The line below is for the first number.



Write each rounded number, then add them.

Answer: _____

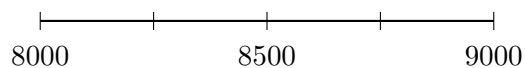
- 7.** Round 5499 two different ways: to the nearest hundred, and to the nearest thousand. The line is marked in thousands.



Why do the two answers point in opposite directions? Write one sentence.

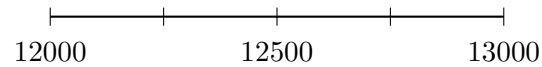
Answer: _____

- 8.** Estimate $8206 - 3891$ by rounding each number to the nearest thousand. Then find the exact difference and say whether the estimate is smaller or larger than it.



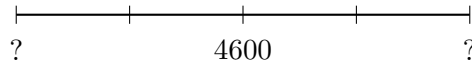
Answer: _____

9. Two hiking trails are measured in metres: the river trail is 12438 metres and the ridge trail is 12509 metres. Round each to the nearest thousand, then compare the two exact lengths with $<$ or $>$. Does the rounded comparison tell you which trail is really longer?



Answer: _____

10. Challenge. A whole number rounds to 4600 when it is rounded to the nearest hundred. Find the least number it could be and the greatest number it could be, then write them in order with $<$ between them.



Answer: _____